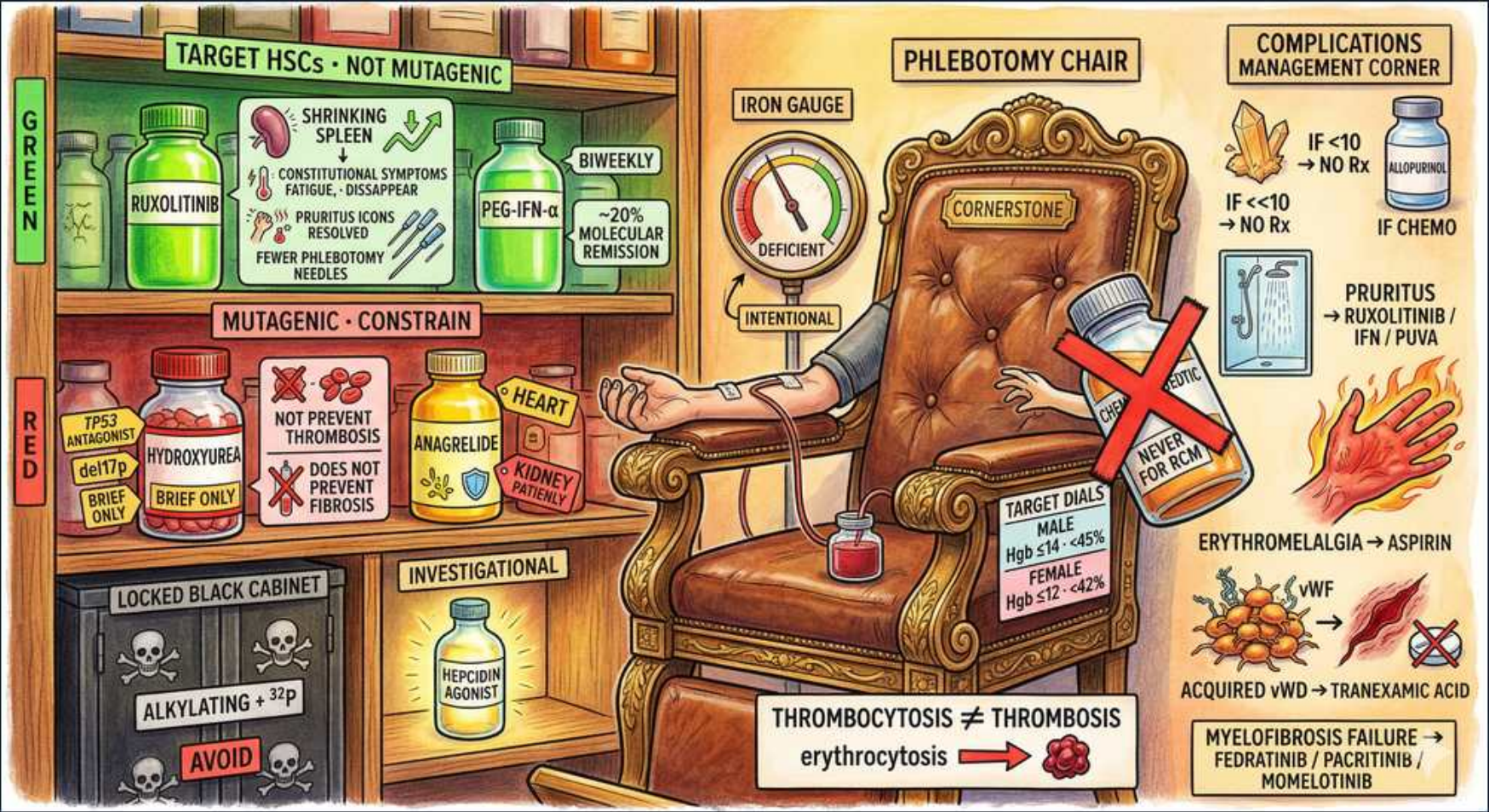


Polycythemia / MPN — Polycythemia Vera treatment



1. Phlebotomy = cornerstone

WHAT YOU SEE

A reclining phlebotomy chair labeled CORNERSTONE with a line draining blood from the patient's arm — bleeding the patient IS the cornerstone therapy.

WHAT IT ENCODES

Therapeutic phlebotomy is the first-line cornerstone for EVERY PV patient regardless of risk, because rapidly lowering the red-cell mass directly cuts hyperviscosity and thrombosis. A typical schedule removes ~250-500 mL initially every few days until the target hematocrit is reached, then as needed for maintenance. It works by inducing iron deficiency, which throttles erythropoiesis. Mnemonic: bleed the patient to thin the blood.

BOARD TRAP

WRONG to reach first for cytoreductive chemotherapy in a newly diagnosed low-risk patient. Discriminator: phlebotomy + low-dose aspirin is the baseline for all PV; cytoreduction (hydroxyurea) is ADDED only for high-risk disease (age >60 or prior thrombosis), not as the initial sole therapy.

2. Hct target <45%

WHAT YOU SEE

TARGET DIALS male Hct <45%, female

WHAT IT ENCODES

Phlebotomize to keep hematocrit BELOW 45% (the same target for men and women per the CYTO-PV randomized trial). Patients kept <45% had roughly a fourfold reduction in cardiovascular death and major thrombosis compared with a <50% target. This single threshold is the most testable number in PV management. Mnemonic: forty-five, stay alive.

BOARD TRAP

WRONG to accept a target of Hct <50% or merely 'within normal range.' Discriminator: CYTO-PV proved <45% is superior to <50%; choosing <50% or a sex-specific normal range is the classic distractor and is associated with more thrombotic events.

3. Iron deficient (intentional)

WHAT YOU SEE

An IRON GAUGE pointing to DEFICIENT marked INTENTIONAL — the needle is deliberately on empty, showing iron deficiency is the wanted effect, not a problem to fix.

WHAT IT ENCODES

Phlebotomy works precisely BY inducing iron deficiency, which limits the substrate for hemoglobin synthesis and restrains red-cell production. Therefore iron supplementation must NOT be given — replacing iron would rebound the hematocrit and undo the therapy. Patients may have low ferritin/microcytosis by design; this is expected and intentional, not a problem to correct.

BOARD TRAP

WRONG to 'correct' the microcytosis or low ferritin in a phlebotomized PV patient with oral/IV iron. Discriminator: the iron deficiency is the intended mechanism of phlebotomy — giving iron causes a hematocrit rebound and increases thrombotic risk, so it is contraindicated.

4. Aspirin (avoid replacing iron)

WHAT YOU SEE

A big red X struck over the box claiming aspirin lowers the hematocrit — the red X means aspirin does NOT thin the red-cell mass.

WHAT IT ENCODES

Low-dose aspirin (81-100 mg daily) is recommended for essentially all PV patients to reduce arterial thrombosis and to relieve erythromelalgia and microvascular symptoms (ECLAP trial). Aspirin controls platelet thromboxane-driven aggregation but does NOT lower the hematocrit, so it complements — never replaces — phlebotomy. Mnemonic: aspirin for the platelets, phlebotomy for the Hct.

BOARD TRAP

WRONG to rely on aspirin to lower the hematocrit, and WRONG to continue aspirin unchanged when platelets are extremely high (>1,000-1,500K). Discriminator: at extreme thrombocytosis, acquired von Willebrand disease makes aspirin a BLEEDING hazard — check vWF activity and hold aspirin/cytoreduce rather than antiplatelet.

5. Hydroxyurea (cytoreduction)

WHAT YOU SEE

RED shelf HYDROXYUREA, NOT PREVENT THROMBOSIS note

WHAT IT ENCODES

Hydroxyurea is the first-line cytoreductive (myelosuppressive) agent for HIGH-RISK PV, defined as age >60 OR a prior thrombotic event. It is a ribonucleotide-reductase inhibitor that lowers red cells, platelets, and white cells. It is added on top of phlebotomy and aspirin, not instead of them. Side effects: cytopenias, oral/leg ulcers, macrocytosis. Mnemonic: high-risk (old or prior clot) → add hydroxyurea.

BOARD TRAP

WRONG to give hydroxyurea to every PV patient or to use it as monotherapy expecting it alone to prevent thrombosis. Discriminator: cytoreduction is reserved for high-risk disease (age >60 or prior thrombosis); low-risk patients are managed with phlebotomy + aspirin alone, and aspirin/phlebotomy remain essential alongside hydroxyurea.

6. Ruxolitinib (JAK inhibitor)

WHAT YOU SEE

GREEN shelf RUXOLITINIB, shrinking spleen

WHAT IT ENCODES

Ruxolitinib is an oral JAK1/JAK2 inhibitor approved as second-line therapy for PV that is resistant to or intolerant of hydroxyurea (RESPONSE trial). It controls hematocrit, shrinks spleen volume, and is especially effective for refractory symptoms including aquagenic pruritus. Watch for cytopenias, reactivation of infections (herpes zoster), and non-melanoma skin cancers.

BOARD TRAP

WRONG to start ruxolitinib as initial cytoreduction before trying hydroxyurea. Discriminator: ruxolitinib is SECOND-line — its niche is hydroxyurea resistance/intolerance, splenomegaly, and refractory symptoms, not first-line cytoreduction in newly diagnosed high-risk PV.

7. PEG-interferon-alfa

WHAT YOU SEE

PEG-IFN- α biweekly, molecular remission

WHAT IT ENCODES

Pegylated interferon-alfa (e.g., ropeginterferon alfa-2b) is a preferred cytoreductive option in YOUNGER patients and in PREGNANCY because it is not leukemogenic/teratogenic and can induce molecular (JAK2 allele-burden) remission. It is dosed roughly every 1-2 weeks. Main drawbacks are flu-like symptoms, mood disturbance/depression, and autoimmune effects. Mnemonic: young or pregnant $\rightarrow$ interferon.

BOARD TRAP

WRONG to default to hydroxyurea in a pregnant or young PV patient needing cytoreduction. Discriminator: interferon is favored in pregnancy and younger patients (non-teratogenic, non-leukemogenic, can reduce JAK2 burden), whereas hydroxyurea is avoided in pregnancy.

8. Avoid alkylating agents

WHAT YOU SEE

LOCKED BLACK CABINET alkylating +32P, AVOID

WHAT IT ENCODES

Alkylating agents (e.g., chlorambucil, busulfan) and radioactive phosphorus (P-32) effectively lower counts but markedly increase the risk of transformation to acute myeloid leukemia, so they are AVOIDED, particularly in patients with long life expectancy. They are reserved only for very elderly patients with no better option. Mnemonic: locked cabinet = leukemogenic, keep it shut.

BOARD TRAP

WRONG to pick an alkylator or P-32 for a young/middle-aged patient to achieve durable count control. Discriminator: these agents are leukemogenic; hydroxyurea, interferon, and ruxolitinib are preferred — alkylators/P-32 are a last resort in the very elderly only.

9. Pruritus management

WHAT YOU SEE

PRURITUS $\rightarrow$ RUXOLITINIB / IFN / PUVA

WHAT IT ENCODES

Aquagenic pruritus is often the most refractory symptom. Stepwise management: antihistamines and avoiding hot showers first, then SSRIs (e.g., paroxetine), then phototherapy (PUVA/narrow-band UVB), with ruxolitinib or interferon for severe/refractory cases that also need cytoreduction. Mnemonic: itch ladder — antihistamine $\rightarrow$ SSRI $\rightarrow$ PUVA $\rightarrow$ ruxolitinib/IFN.

BOARD TRAP

WRONG to treat PV pruritus only with topical emollients/steroids and declare it untreatable when those fail. Discriminator: PV itch is mediated by the clonal mast-cell/basophil mass, so systemic agents (SSRIs, PUVA, ruxolitinib, interferon) — not topicals — are the effective options.

10. Erythromelalgia $\rightarrow$ aspirin

WHAT YOU SEE

Red hand ERYTHROMELALGIA = ASPIRIN

WHAT IT ENCODES

Erythromelalgia (burning red painful extremities) responds rapidly — often within hours to a day — to low-dose aspirin, because the pathology is platelet/thromboxane-mediated microvascular thrombosis. This dramatic aspirin response is both diagnostic and therapeutic. Mnemonic: burning red feet that vanish with aspirin = MPN.

BOARD TRAP

WRONG to escalate to opioids, gabapentin, or vasodilators for erythromelalgia before trying aspirin. Discriminator: the rapid relief with low-dose aspirin distinguishes MPN-related erythromelalgia from other causes of painful red extremities, so aspirin is the first move.

11. Acquired vWD $\rightarrow$ tranexamic

WHAT YOU SEE

vWF ACQUIRED vWD $\rightarrow$ TRANEXAMIC ACID

WHAT IT ENCODES

With extreme thrombocytosis (platelets $>1,000$ - $1,500K$), large vWF multimers are consumed, producing acquired von Willebrand disease and mucocutaneous bleeding. Management: HOLD aspirin, lower the platelet count with cytoreduction, and use tranexamic acid for bleeding (desmopressin gives transient benefit). Always check vWF activity before adding aspirin at very high platelet counts.

BOARD TRAP

WRONG to add or continue aspirin in a PV patient with bleeding and platelets $>1,000K$ assuming high platelets mean high clot risk. Discriminator: extreme thrombocytosis causes acquired vWD and a BLEEDING tendency — aspirin worsens it; the correct steps are to lower platelets and give tranexamic acid.

12. Post-PV myelofibrosis tx

WHAT YOU SEE

MYELOFIBROSIS FAILURE: fedratinib/pacritinib/momelotinib

WHAT IT ENCODES

Progression to post-PV myelofibrosis (marrow failure, cytopenias, massive splenomegaly, constitutional symptoms) is treated with JAK inhibitors: ruxolitinib or fedratinib, with pacritinib useful when platelets are very low ($<50K$) and momelotinib favored when anemia is prominent. Allogeneic stem-cell transplant is the only curative option in eligible higher-risk patients. Mnemonic: spent-phase PV $\rightarrow$ JAK inhibitor (transplant if able).

BOARD TRAP

WRONG to continue phlebotomy-based management once a PV patient has transitioned to the spent/fibrotic phase with cytopenias. Discriminator: in post-PV myelofibrosis the problem is marrow failure (low counts), so therapy shifts to JAK inhibitors/transplant — phlebotomy is no longer appropriate.

13. Thrombocytosis ≠ thrombosis

WHAT YOU SEE

THROMBOCYTOSIS ≠ THROMBOSIS, erythrocytosis → clot

WHAT IT ENCODES

In PV, thrombotic risk is driven by the ERYTHROCYTOSIS (red-cell mass / hematocrit), NOT by the platelet count. This is why the management goal is hematocrit <45% via phlebotomy rather than platelet normalization. Paradoxically, extreme platelet elevation increases BLEEDING (acquired vWD), not clotting. Mnemonic: red cells clot, sky-high platelets bleed.

BOARD TRAP

WRONG to treat PV by chasing a normal platelet count to prevent thrombosis. Discriminator: lowering the hematocrit below 45% is what reduces thrombosis; targeting the platelet count misses the mechanism, and pushing antiplatelet therapy at extreme thrombocytosis can precipitate acquired-vWD bleeding.
